Saksham Srivastava

DevOps / Platform Engineer | Kubernetes | Observability (OpenTelemetry) | Bare-metal networking

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SUMMARY

Platform engineer running 7 Kubernetes clusters (bare-metal + EKS, ~80 nodes, ~160 services) on a 3-person team covering DevOps, SRE, networking and FinOps. Own observability end to end, from a custom OTel agent extension to root-causing production collapses from traces. Merged upstream contributions to OpenTelemetry and fluent-bit; first-author Springer paper.

EXPERIENCE

DevOps Engineer, Enveu Aug 2025 - Present | Remote

OTT / video SaaS platform, mostly Java microservices on Kubernetes and AWS

Observability & performance

- **Wrote an OpenTelemetry Java agent extension** (open source) that drops a noisy root-span pattern and its entire child tree before export. One INTERNAL span was producing ~60 GB of traces every 6 hours; the same volume now takes 4-5 days.
- **Rolled out the OTel kube-stack with ECK / Kibana / APM:** operator-based agent injection, daemonset collectors, *elasticapm* connector, span-drop rules and scrape cleanup.
- **Led a 600-VU load test and post-mortem** that found congestion collapse at 26% of target load (p50 5 ms to 1.4 s in 4 min). Traced it to uninstrumented HikariCP connection-acquisition waits (~1.3 s unattributed per request), not the database; also surfaced HPA ceilings and 32x span amplification.
- **Diagnosed an Elasticsearch translog runaway** (24 GB translog vs 7.4 GB segments) by elimination; root cause was ~4 MB indexing buffer per shard across 49 shards. Captured as a runbook.

Kubernetes, networking & migrations

- **Bare-metal K8s** (plus 5 EKS clusters on Karpenter spot): L4 load balancing, ACME TLS, and a fix for broken hairpin TCP (LB sent no PROXY protocol header) via CoreDNS rewrites that resolve ingress hosts to in-cluster services.
- **Architected a zero-downtime AMD to ARM migration** company-wide: multi-arch pipelines and rollout process; 20 services cut over, 80+ queued on the same path.
- Led Docker to Kubernetes (QA) and NGINX to Envoy ingress migrations; stood up a production S3-backed Harbor registry with replication, RBAC and scanning.
- Deployed a 7-component Java platform onto bare-metal K8s in under 6 hours with zero documentation, reverse-engineering dependencies and configs.

Security & incidents

- **First responder on an AWS access-key compromise:** rotated credentials, scoped blast radius; now driving the org-wide move from hardcoded keys to IRSA (incl. cross-account Route 53 role chaining). Resolved a 4-day unalerted webhook outage.

DevOps Intern, Careers360 Dec 2024 - Aug 2025

- **Right-sized over-provisioned infrastructure, cutting cost ~40% with zero downtime.**
- Set up Velero-based Kubernetes backups for cluster resources and persistent volumes.

OPEN SOURCE & PROJECTS

- **opentelemetry-helm-charts #2377** (merged): kube-stack schema rejected *instrumentation.java.extensions*, blocking Java agent extensions via Helm.
- **fluent/fluent-bit** (merged): install script silently broke on Debian/Ubuntu after the package host began 302-ing to S3.
- **otel-pattern-sampler**: the span-dropping extension above, built on AutoConfigurationCustomizerProvider with a parent-based sampler decorator. MIT.
- **cyclops** (in progress): Go/kubebuilder controller that emails a daily report of cert-manager certificates that failed to renew, with cert-manager's own failure diagnostics. Design documented in 13 ADRs.

PUBLICATIONS

Investigating Cultural Bias: A Comparative Study of Large Language Models. S. Srivastava, P. Singh, S. Khetarpaul, U. Sharma. *PReMI 2025, Springer LNCS vol. 16357*, pp. 425-433 (2026). doi:10.1007/978-3-032-18477-1_45

Benchmarking LLM powered Chatbots: Methods and Metrics. D. Banerjee, P. Singh, A. Avadhanam, S. Srivastava. *arXiv:2308.04624* (2023).

SKILLS

Containers & K8s	Kubernetes (bare-metal, EKS), Helm, Karpenter, Envoy, NGINX, CoreDNS, Harbor, Velero, Docker, kubebuilder
Observability	OpenTelemetry (SDK, Java agent, Collector, Operator), Elasticsearch / ECK, Kibana, APM, SigNoz, Prometheus
Cloud & IaC	AWS (EKS, IAM / IRSA, Route 53, RDS, S3), Terraform
Languages & core	Go (learning), Java, Python, Bash, SQL, Linux, TCP/IP, DNS, L4/L7 load balancing, TLS

EDUCATION

B.Tech, Computer Science and Engineering, Shiv Nadar Institution of Eminence, Delhi NCR 2021 - 2025